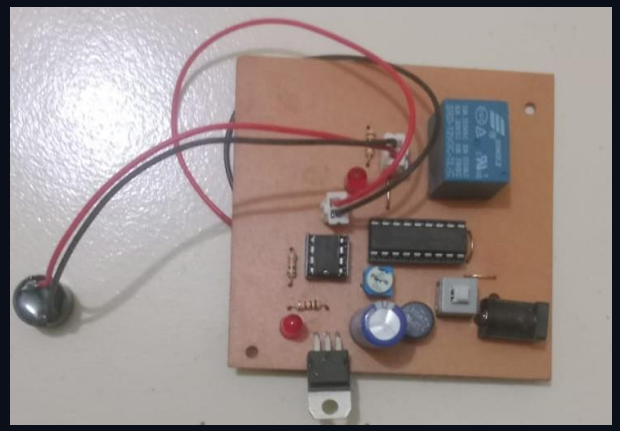


DIGITAL ELECTRONICS PROJECT FOR INNOVATIVE

:

Digital Electronics (UEC3001)



Fire Detection and Automatic Alarm System

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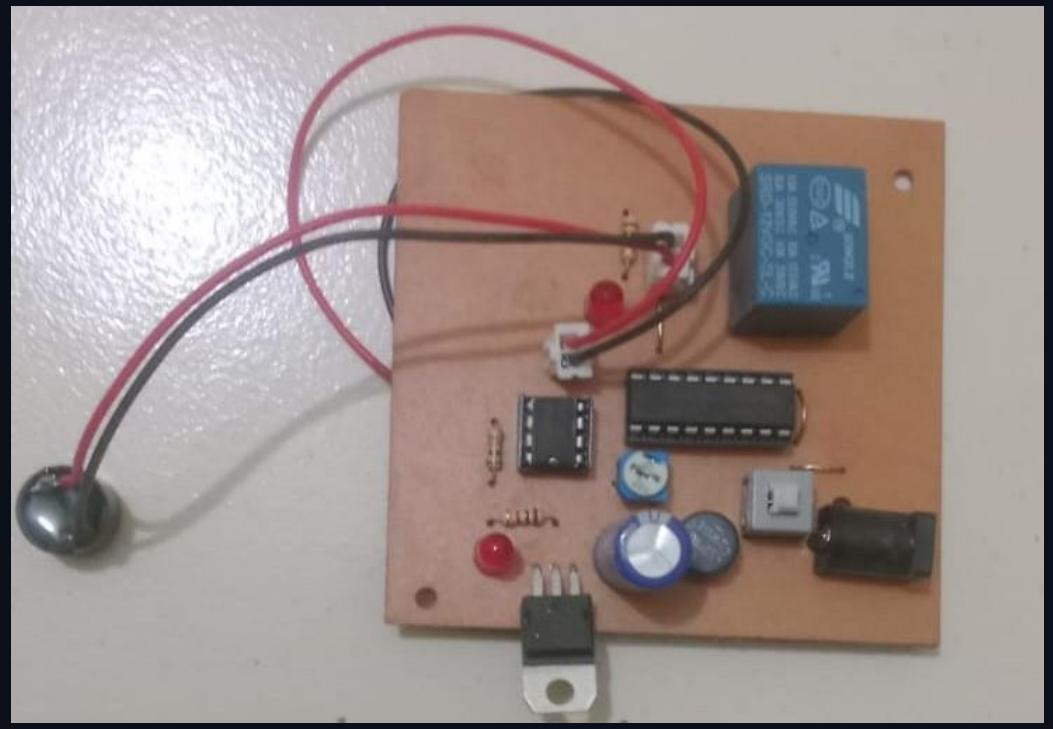
Prabhat Srivastava

Introduction

A Fire Detection and Automatic Alarm System is designed to detect the presence of fire and provide an immediate warning.

This project uses a Flame Sensor to detect infrared radiation emitted by a flame. The sensor generates an electrical signal that controls a Transistor-based Switching Circuit, which activates a Buzzer to produce an audible alert.

The system demonstrates the practical application of electronic sensors, signal processing and transistor switching for basic fire detection.

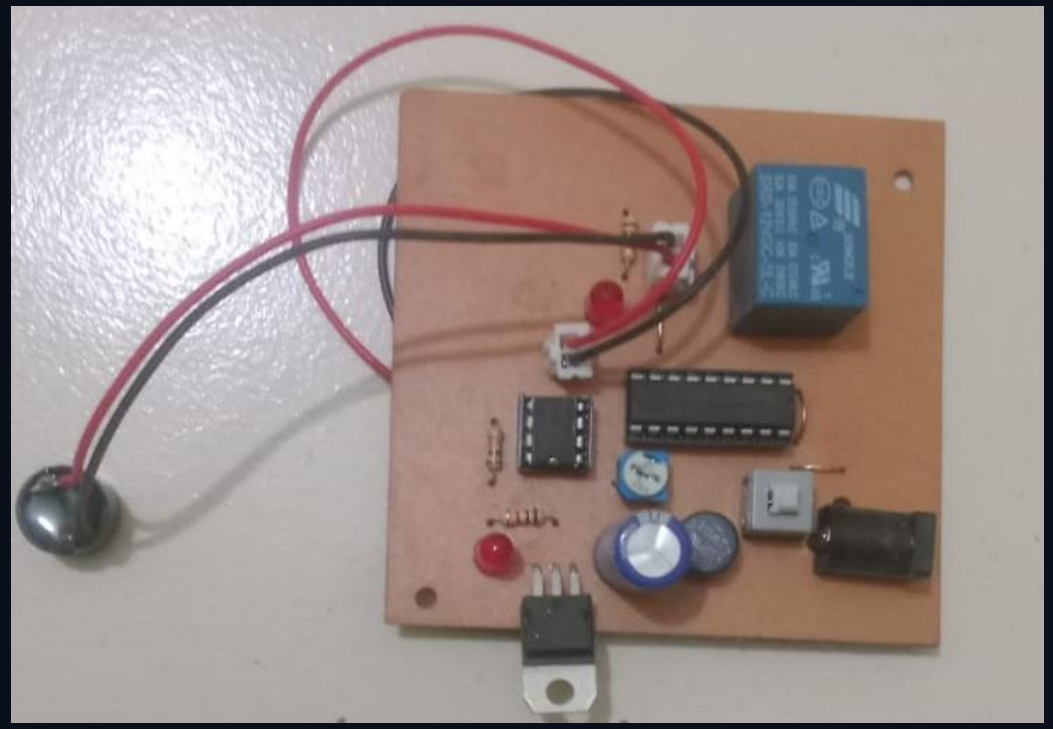


Problem Statement

Fire accidents can cause serious damage to human life and property. In many situations, delayed fire detection due to manual monitoring can increase the risk of accidents.

Therefore, there is a need for a simple automatic system that can detect the presence of a flame and generate an immediate warning.

This project aims to develop a flame sensor-based system that uses a transistor switching circuit to activate a buzzer automatically when a flame is detected.



Objectives & System Block Diagram system

Objectives of the Project
- To detect the presence of fire using a flame sensor.
- To convert the detected flame condition into an electrical signal.
- To use a transistor as an electronic switching device.



1. Instant Flame Sensing

Detect 760–1100 nm infrared radiation reliably.

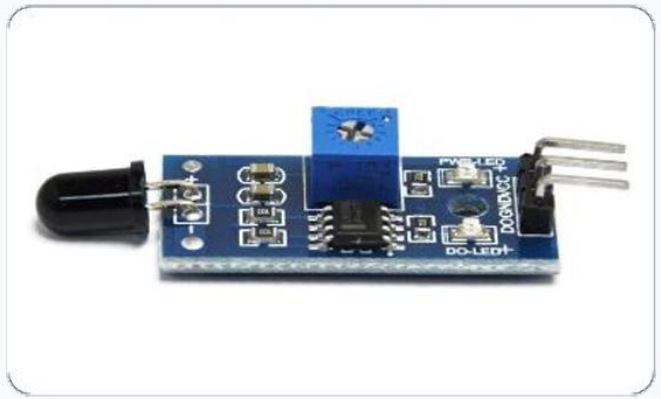
2. Armed Gate Control

Validate flame signal with system enable switch.

3. Dual Audio-Visual Alert

Drive piezoelectric buzzer and LED simultaneously.

Components Required





IR Flame Sensor (YG1006)

Outputs 0V (LOW) upon IR detection.

IC 7404 (Hex NOT Gate)

Inverts active-LOW sensor signal to HIGH.

IC 7408 (Quad AND Gate)

Combines inverted flame signal with enable switch.

5V Buzzer & Red LED

Provide immediate audible and visual warnings.

330Ω & 10kΩ Resistors

Limit LED current and pull-down enable input.

Breadboard & +5V DC Supply

Solderless prototyping base with regulated TTL power.

Digital Electronics Concepts &Logic Gates

TTL Logic Levels: LOW (0) = 0V–0.8V | HIGH (1) = 2.0V–5.0V

System Boolean: $Y = \overline{F} \cdot E$

NOT Gate (IC 7404)

Expression: $X = \overline{A}$

Input A	Output X
1 (Flame)	1
2 (Safe)	0

AND Gate (IC 7408)

Expression: $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

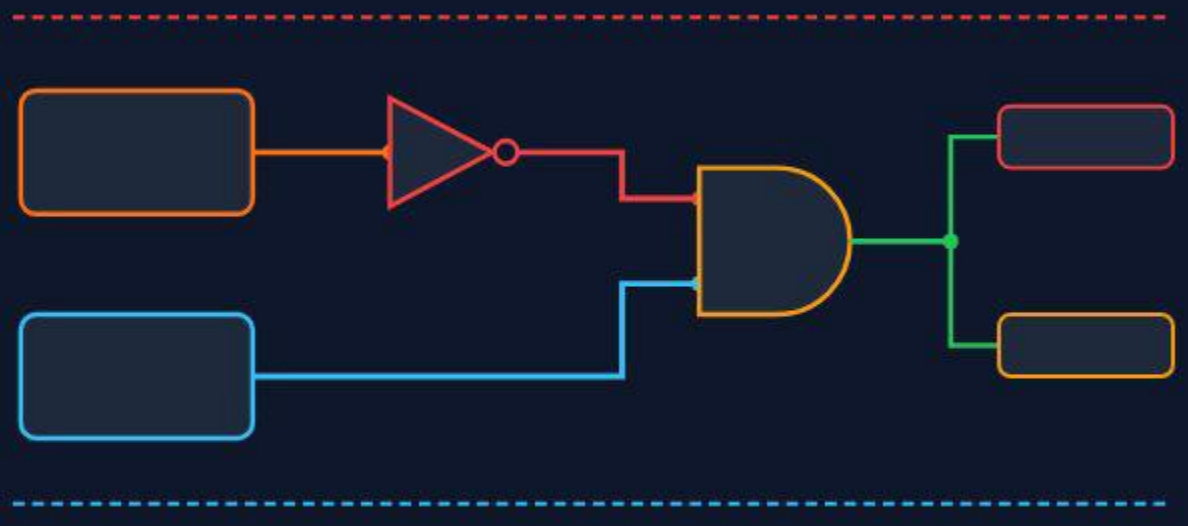
OR Gate (IC 7432 Concept)

Multi-Zone: $Z = A + B$

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

Logic Circuit Diagram & System Truth Table

PINOUT: IC 7404 & IC 7408 (Pin 14 = +5V Vcc, Pin 7 = GND)



Sensor D0 → 7404 Pin 1

7404 Pin 2 → 7408 Pin 1

Enable → 7408 Pin 2

7408 Pin 3 → LED & Buzzer

BOOLEAN OUTPUT EQUATION

$$Y = \overline{F} \cdot E$$

Flame (F)	7404 ($\overline{F}$)	Enable (E)	Alarm (Y)
1 (No)	0	0 (OFF)	0 (OFF)
1 (No)	0	1 (ON)	0 (OFF)
0 (Fire)	1	0 (OFF)	0 (Muted)
0 (Fire)	**1**	**1 (ON)**	**1 (ALARM)**

Step-by-Step Working Principle

01

Normal Standby

No IR radiation keeps sensor output HIGH 1

02

Flame Detection

Photodiode drops sensor D0 to logic LOW 0

03

Signal Inversion

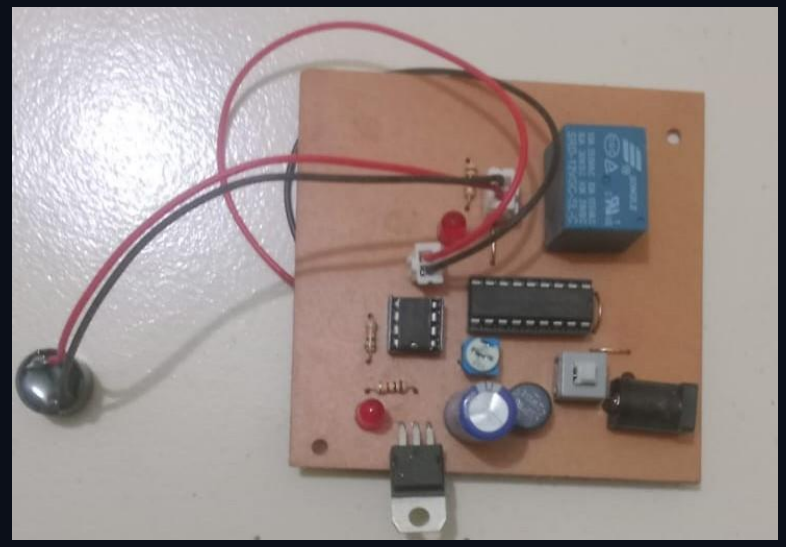
IC 7404 inverts LOW input into HIGH 1

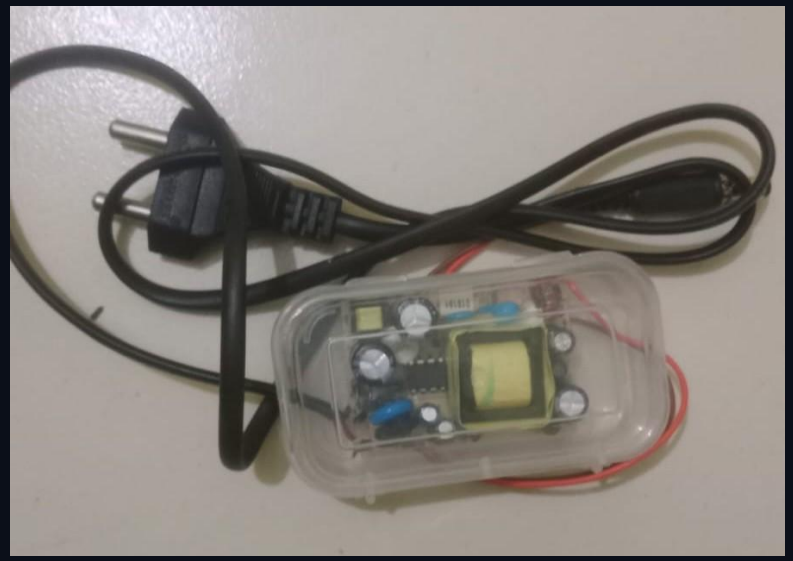
04

Alarm Activation

IC 7408 outputs HIGH (1) to trigger buzzer.

Breadboard Working Model & Applications





Key Applications

Chemistry laboratories, server rooms, kitchens, and industrial warehouses.

Core Advantages

Zero firmware bugs, nanosecond propagation, low hardware cost.

Current Limitations

Requires direct line-of-sight; lacks remote wireless alerts.

Conclusion & Proposed Future Scope

CURRENT HARDWARE IMPLEMENTATION

The project successfully detects flames using a flame sensor and activates a buzzer through a transistor switching circuit. The working prototype demonstrates a simple and effective approach to automatic flame detection and alert generation.

PROPOSED FUTURE ENHANCEMENTS

- Smoke Sensor Integration
- Temperature Monitoring
- GSM-Based SMS Alerts
- IoT-Based Remote Monitoring
- Battery Backup

GitHub Repository:
https://github.com/adarshhpandey12/Fire-Detection-and-Automatic-Alarm-System

Repository Name: Fire-Detection-and-Automatic-Alarm-System